

(Soumansu Sekhar Nayak  
2141014141)

## Assignment-2 [LA network]

1. parallel computing Example - weather forecasting systems

Application:-

National weather service and organizations like NASA use super computers that run parallel computing models to simulate and predict weather patterns.

How it works:-

- The atmosphere is divided into 3D grids.
- Each grid is assigned to processor or core which simultaneously computes temperatures, wind, pressure etc.
- This results are then combined to produce a global forecast.

Why parallel computing is important:-

- Weather data changes rapidly real-time prediction require massive computation.

2. Networked System Eg:- Online Multiplayer Gaming.

Games like PUBG, Call of Duty or Fortnite use client server architecture with networked system to enable real time multiplayer game play.

How it works:-

- Game clients (players devices) communicate with central game servers over the internet.
- Servers handle logic like player position, movement, actions and environment updates.
- All devices are synchronized via a reliable, low latency network.

• Why networked system are important:-

- Enables real-time interaction among millions of player world wide
- keeps gameplay consistent and fair.
- Supports cloud-based scaling (e.g. more servers during high traffic).